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10 October 2026

Dr Daniel Lee
Department of Chemical Engineering, School of Engineering
The University of Manchester
Manchester M13 9PL, United Kingdom

Re: Research Associate in Solid-State NMR Spectroscopy, ref. SAE-032399

Dear Dr Lee,

I am applying for the position of Research Associate in Solid-State NMR Spectroscopy (ref. SAE-032399) on the EPSRC Programme Grant on Molecular Solid Solutions. I completed my PhD in Chemical Sciences at the Institute of Chemistry, UNAM (Mexico), in 2026 with Prof. Braulio Rodríguez-Molina, and the degree was awarded with honours. My thesis used solid-state NMR with diffraction to relate molecular motion in organic and metal-organic crystals to their macroscopic behaviour.

The chapter closest to this project concerns a solid solution. I synthesised UiO-68-type zirconium MOFs containing a terphenyl rotor linker and a longer azobenzene photoswitch linker in nominal ratios from 10 to 70%. Because the linkers differ in length, the PXRD reflections, which are dominated by the Zr_6 clusters, shifted to lower angle as the photoswitch content increased. I built periodic models with the linkers in ordered positions, and the calculated patterns of the DFT-optimised models contained extra reflections that were absent experimentally, which pointed to a random distribution of linkers within one phase. ^{13}C CP-MAS confirmed that both linkers were incorporated. With a deuterated rotor linker I recorded ^{2}H spectra between 178 and 338 K, which were fitted with 90° and 180° jumps above 10^{7} Hz using electric field gradients calculated in CASTEP. After loading hexafluorobenzene, which I confirmed by ^{19}F MAS and quantified by ^{19}F NMR with standard additions, the 180° jumps were suppressed and the 90° site populations depended on the photoswitch state (ACS Mater. Lett., 2026). In molecular solid solutions of similar-sized components, diffraction changes little with composition, and the local environment seen by solid-state NMR becomes the main source of information.

In a toluene solvate of a pyrrolo[3,2-b]pyrrole that jumps when cooled to 77 K (Mater. Chem. Front., 2026), I prepared the toluene-d_8 analogue and simulated in NMR WebLab the ^{2}H lineshapes recorded from 178 to 338 K, using quadrupolar parameters from GIPAW calculations, which showed methyl rotation and 180° ring flips above 10^{7} Hz. Variable-temperature SCXRD gave a thermal expansion coefficient of 271 MK^{-1} oblique to the face in contact with the cold surface, and we attributed the jump to this anisotropic expansion (with Dr Martin Dračínský and Dr Jan Blahut, IOCB Prague). I was responsible for the NMR analysis, in solution and in the solid state, in all 14 of my publications. For example, I interpreted and modelled the ^{2}H spectra of acetone and tetrahydrofuran in mixed-solvate cocrystals (Angew. Chem. Int. Ed., 2026) and of a guest-loaded zirconium MOF (Chem. Sci., 2020), and interpreted the ^{13}C CP-MAS data of a thermosalient crystal (Chem. Mater., 2025).

I ran my own experiments on a Bruker 500 MHz spectrometer, from acquisition to interpretation and modelling, and also acquired ^{31}P and ^{15}N CP-MAS spectra and analysed ^{13}C T_1 and solid-state ^{1}H and ^{19}F relaxometry data. I made the compounds in my thesis, including the deuterated analogues, and characterised them by SCXRD, PXRD, DSC and TGA. I have not yet used DNP. The UK DNP MAS NMR Facility in Manchester, and the chance to extend this work to ^{17}O and natural-abundance ^{15}N, are among my main reasons for applying.

I co-supervised four bachelor's theses and nine research placements and managed purchasing and inventory for our laboratory. I currently work at the Mexican Institute of Industrial Property until December 2026 and could start in January 2027. Thank you for considering my application.

Yours sincerely,

Dazaet Galicia-Badillo, PhD